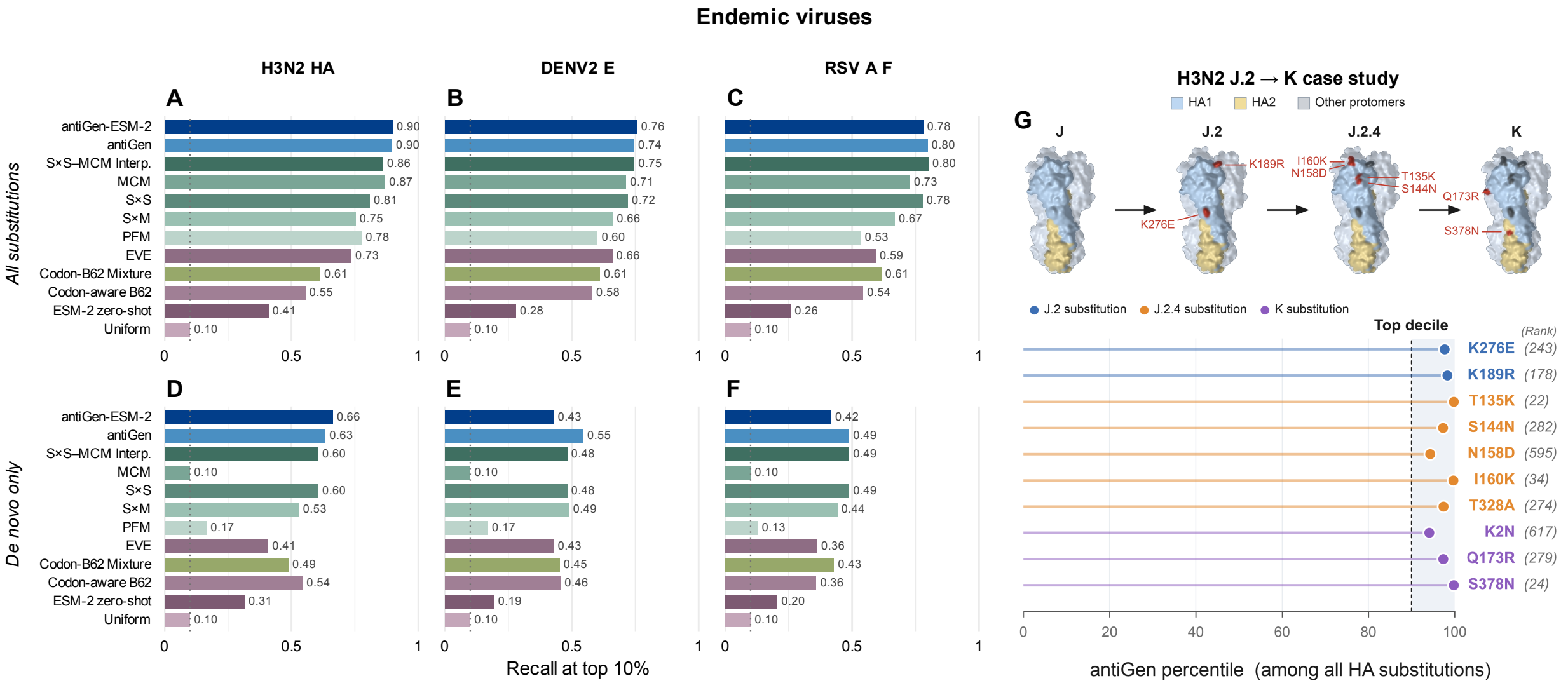

Endemic viruses
H3N2 HA
DENV2 E
RSV A F
A
B
C
D
E
F
All substitutions
De novo only
antiGen-ESM-2
antiGen
S×S–MCM Interp.
MCM
S×S
S×M
PFM
EVE
Codon-B62 Mixture
Codon-aware B62
ESM-2 zero-shot
Uniform
0.90 0.90 0.86 0.87 0.81 0.75 0.78 0.73 0.61 0.55 0.41 0.10
0.76 0.74 0.75 0.71 0.72 0.66 0.60 0.66 0.61 0.58 0.28 0.10
0.78 0.80 0.80 0.73 0.78 0.67 0.53 0.59 0.61 0.54 0.26 0.10
0.66 0.63 0.60 0.10 0.60 0.53 0.17 0.41 0.49 0.54 0.31 0.10
0.43 0.55 0.48 0.10 0.48 0.49 0.17 0.43 0.45 0.46 0.19 0.10
0.42 0.49 0.49 0.10 0.49 0.44 0.13 0.36 0.43 0.36 0.20 0.10
0 0.5 1
Recall at top 10%
G
H3N2 J.2 → K case study
HA1
HA2
Other protomers
J
J.2
J.2.4
K
K189R
K276E
I160K
N158D
T135K
S144N
Q173R
S378N
J.2 substitution
J.2.4 substitution
K substitution
Top decile
(Rank)
K276E (243)
K189R (178)
T135K (22)
S144N (282)
N158D (595)
I160K (34)
T328A (274)
K2N (617)
Q173R (279)
S378N (24)
0 20 40 60 80 100
antiGen percentile (among all HA substitutions)